Module 1: Introduction

Andrea Moglia, Marco Paracchini

Laboratory of Nearable Technologies for Health Data Science
A.Y. 2025/2026

About Us

Andrea Moglia

He received his Master's degree in Mechanical Engineering at the University of Brescia (Italy) and the PhD in Microsystems Engineering at the University of Rome Tor Vergata (Italy). He participated in research projects on robot-assisted surgery, funded by the Department of Defense of the United States, and capsule endoscopy, supported by the South Korea government. He is working on artificial intelligence in medical imaging and surgery focused on multi-modal large language models.

Marco Paracchini

He obtained a Master's degree in Mathematical Engineering in 2014 and a PhD in Information Technology in 2020 both from Politecnico di Milano. He worked on many topics related to signal processing, computer vision and machine learning including: SLAM, visual search, rPPG, tomography and eye tracking. He was part of two H2020 European projects and he collaborated with many companies such as STMicroelectronics, General Motors, Univision and Xnext. He is currently involved in a Politecnico and Luxottica joint project developing eye tracking and SLAM for smart eyewear (SEL).

Course Objectives

Bloom Taxonomy

- Cognitive skills (knowledge);

- Technical skills (psychomotor);

- Affective skills (e.g., communication, team training).

Source: https://teaching.uic.edu/cate-teaching-guides/syllabus-course-design/blooms-taxonomy-of-educational-objectives/

Healthcare

Definition (Merriam-Webster Dictionary):

"Efforts made to maintain, restore, or promote someone's physical, mental, or emotional well-being especially when performed by trained and licensed professionals"

The Healthcare Landscape

Clinical and Medical Care

- Primary Care: General practitioners, family medicine, pediatrics
- Specialized Care: Cardiology, oncology, neurology, etc
- Emergency and Critical Care: ER, ICU, trauma management
- Surgical Care: General surgery, orthopedics, MIS procedures

Public Health and Preventive Medicine

- Epidemiology and disease surveillance
- Health promotion and wellness programs
- Vaccination and immunization programs
- Environmental and occupational health

Pharmaceutical and Biotechnology

- Drug discovery and development
- Pharmacovigilance and regulatory affairs
- Personalized medicine and genomics

The Healthcare Landscape

Nursing and Allied Health Professions

- Nursing (ICU, pediatric, geriatric, etc.)
- Physiotherapy, occupational therapy, speech therapy
- Radiology, laboratory, and pathology services

Mental and Behavioral Health

- Psychiatry, psychology, counseling
- Substance abuse treatment
- Neurodevelopmental and cognitive therapy

Health Insurance and Healthcare Policy

- Public health insurance (Medicare, Medicaid, NHS, etc.)
- Private health insurance and reimbursement models
- Healthcare policy and ethics

The Healthcare Landscape

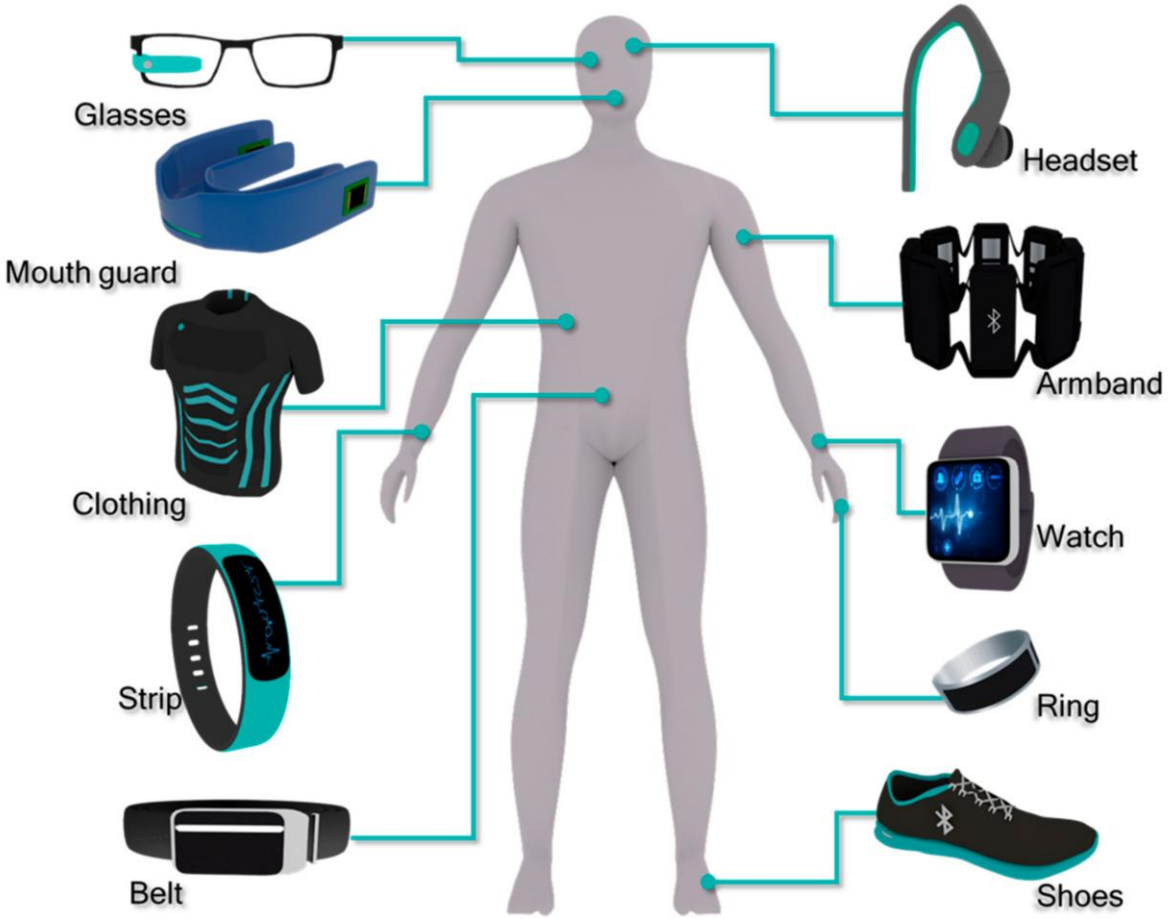
Hospital and Healthcare Management
- Healthcare operations and administration
- Quality and safety in patient care
- Medical supply chain and logistics

Alternative and Integrative Medicine
- Traditional medicine (Ayurveda, Chinese medicine)
- Chiropractic and acupuncture
- Holistic and wellness-based treatments

Medical Research and Education
- Clinical trials and evidence-based medicine
- Medical education and training
- Bioethics and legal aspects of healthcare

Health Informatics and Digital Health
- Electronic Health Records (EHRs) and Health Information Systems
- Telemedicine and remote patient monitoring
- Artificial Intelligence and Big Data in healthcare
- **Wearables** and **nearable** devices for health tracking

Wearable Devices



Nearables (Nearable Technology)

- Initially described as smart, connected objects that broadcast data about their location, motion and temperature;
- Type of smart objects invented by Estimote Inc. in 2014 as a part of a marketing campaign.

Source: Bianchi MT. Metabolism 2018;84:99-108

Term "Nearable" inspired by:

Pebble Watch
(by Pebble Technology Corporation)
Acquired by Fitbit

Google Glass

Nearables (Nearable Technology)

"Consumer tracking devices that are <u>not actually worn</u> in contrast with wearable"

Source: Matt T, et al. Metabolism 2018;84:99-108

"Everyday sensorized objects <u>with transmitting capabilities</u> such as smartphones"

Source: Di Rienzo M, et al. Sensors 2021;21,1291

"Smartphone: accelerometer, gyroscope, and camera"

Nearables in Healthcare

Their main advantage is that they can measure signals and information in a non invasive manner.

In healthcare nearable devices monitor biological signals and information from a distance without direct contact with individuals.

Some healthcare applications:

- Measure heart rate;
- Track thoracic movements associated with respiration, yielding a time series signal similar to what one would obtain from a chest or abdomen belt;
- Body motion, pressure, blood volume, temperature, and biochemical markers.

Source: Bianchi MT. Metabolism 2018;84:99-108

Source: Di Rienzo M, et al. Sensors 2021;21,1291

Motion Sensors

Wearables

Accelerometers
- Measure acceleration in multiple directions (X, Y, Z axes);
- Used for fall detection, activity tracking, and movement analysis;
- Found in smart patches, wearables for elderly care.

Gyroscopes
- Detect angular velocity (rotation and orientation);
- Help in gait analysis, posture monitoring, and dizziness detection;
- Used in rehabilitation and physical therapy applications.

Pressure Sensors
- Detect changes in pressure from movement or force;
- Used in smart insoles, bed occupancy sensors monitoring.

Nearables

Infrared (IR) Sensors
- Detect proximity and movement using infrared light;
- Inexpensive and work also in the dark;
- Affected by the surface properties (texture and color) of the object;
- Motion tracking for patient monitoring.

Time-of-Flight (ToF) Sensors
- High precision and suitable for real-time application;
- Work also in the darkness;
- Expensive and short range (about 5 m);
- Motion tracking for patient monitoring.

Ultrasound
- Inexpensive and higher range;
- Affected by temperature, humidity, and other environmental conditions;
- Less precise and slower than ToF;
- Useful for detecting breathing patterns and heartbeats.

Optical & LiDAR Sensors
- Use laser or light-based detection for gesture control and spatial tracking;
- Patient monitoring, fall prevention, and hospital automation.

Temperature Sensors

Wearables

Thermistors
- ❖ Resistive temperature sensors that change resistance based on heat;
- ❖ Fast response to temperature change;
- ❖ Found in wearable patches for fever monitoring and smart thermometers.

Resistance Temperature Detectors (RTDs)
- ❖ Measuring a very broad range (100°C or higher);
- ❖ Less accurate than thermistors.

Thermocouples
- ❖ Detect temperature changes based on voltage differences between two metals;
- ❖ High-precision;
- ❖ Example: monitoring hypothermia.

Nearables

Infrared (IR) Temperature Sensors
- ❖ Measure temperature without direct contact by detecting infrared radiation;
- ❖ Used in non-contact thermometers, fever screening, and room temperature monitoring.

Common Applications
- ❖ Fever Monitoring;
- ❖ Smart Hospitals: Room and bed temperature regulation;
- ❖ Neonatal & Elderly Care: Remote monitoring for at-risk patients.
- ❖ Smart Wearables: fitness trackers.

Benefits of Nearables in Health

They can help transform healthcare from the current reactive, one-size-fits-all, hospital-centered, and volume-based system into a future proactive, personalized, decentralized, and valued-based system.

Nearables in Health Data Science

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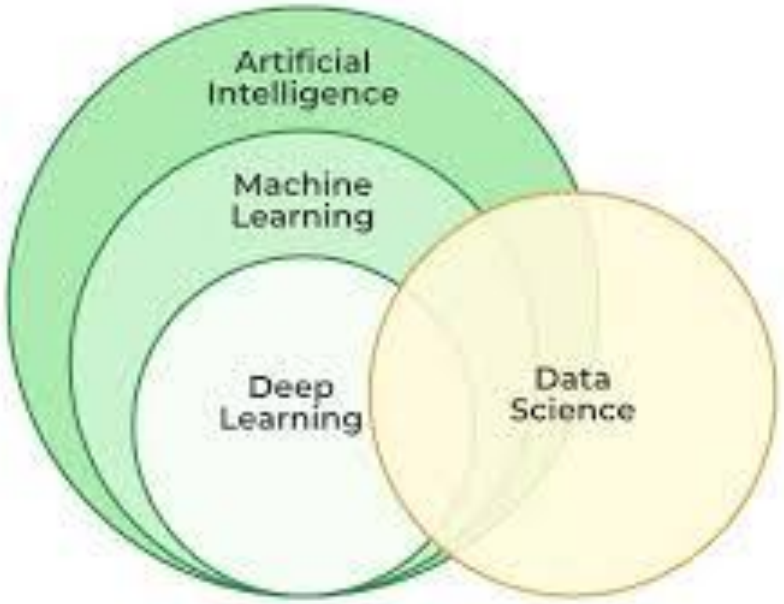
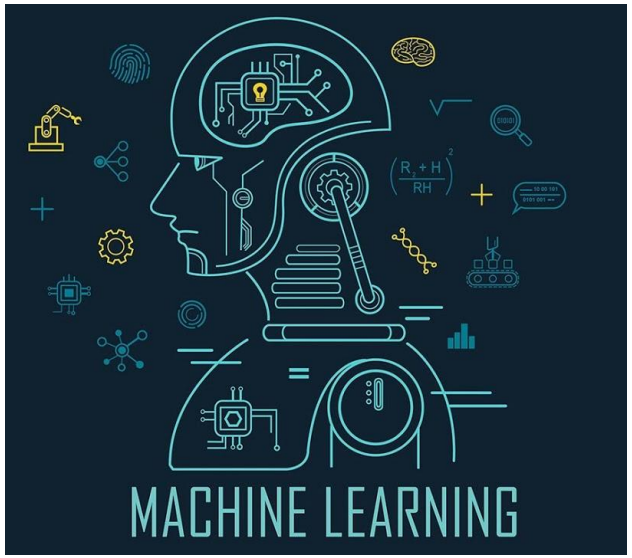


Nearables in Health Data Science

This course aims to provide foundational knowledge for developing health data science applications, with a focus on data from wearable and nearable sensors and image processing.



Laboratory Activities

Projects (3-4 students each)

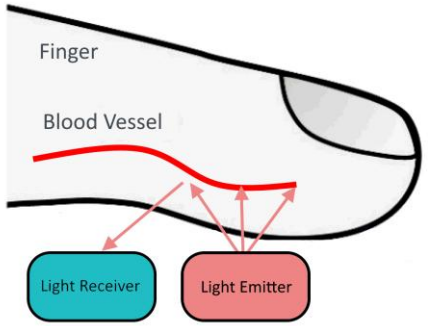
- Ground truth estimation (Wearable)
- Development of Machine Learning/ Deep learning models (Nearable);
- Design of graphical user interface;
- Creation of GitHub repository with projects file.

Evaluation

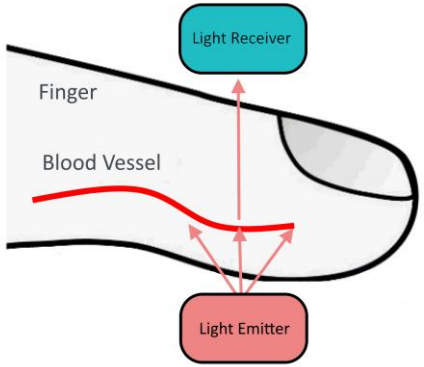
- 10-minute presentation + demo;
- GitHub repository (https://n-health-lab.github.io/).

Projects Proposals - rPPG

PhotoPlethysmoGraphy-PPG working principals



Using reflected light

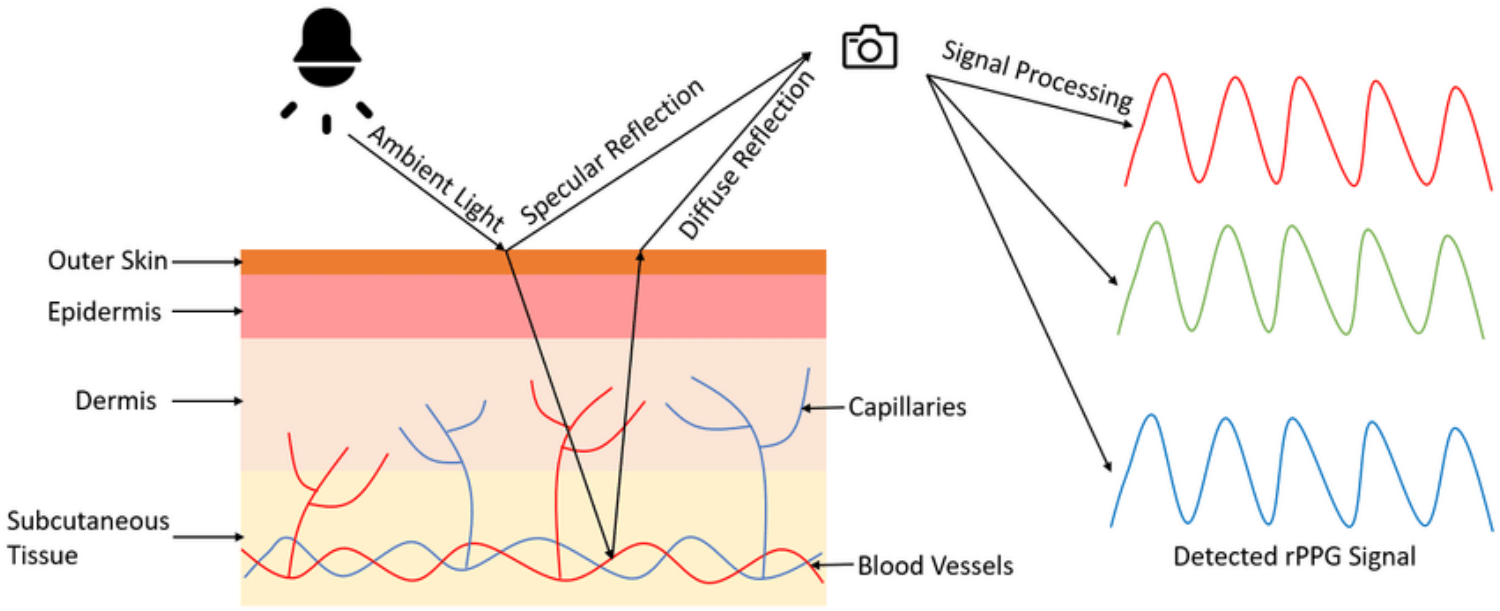


Using transmitted light

- When bloods flows into vessel it changes the reflective and transmitted coefficients
- By studying the dynamic of transmitted or reflected light intensity heart related features could be extracted: heart rate (HR), heart rate variability (HRV), respiratory rate etc.
- Using two different light emitters (red and infrared light) oxigenation ratio could be estimated (pulsoximeter)

Projects Proposals - rPPG

Remote PhotoPlethysmoGraphy-rPPG working principals

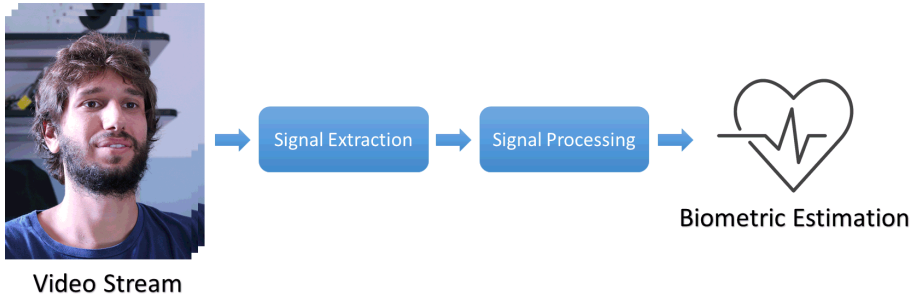


Using reflected light

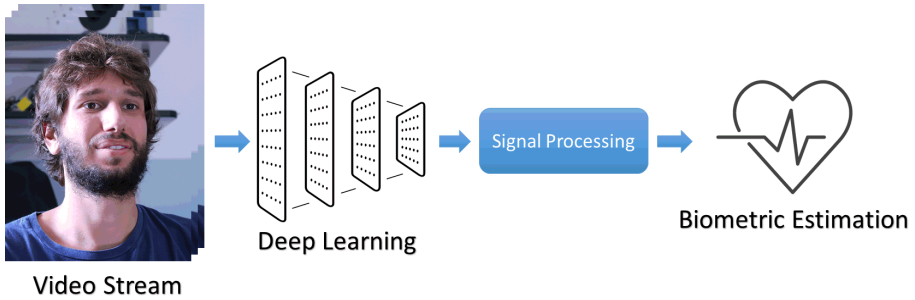
- Same principle as before
- Ambient or controlled light could be used
- Received lisght is a mix between: specular reflection (no PPG signal), diffusive reflection (PPG signal!), environment light and noise
- Signal is much more noisy!

Projects Proposals - rPPG

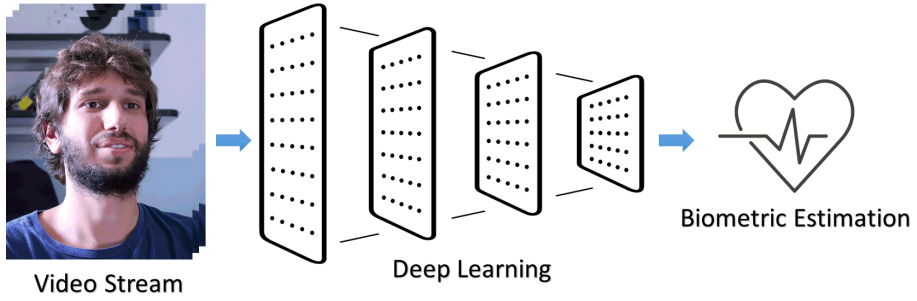
The use of deep learning in rPPG pipeline



Traditional approach
Signal processing could be used both for signal extraction (face detection) + signal processing (filtering, PCA, ICA, FFT)



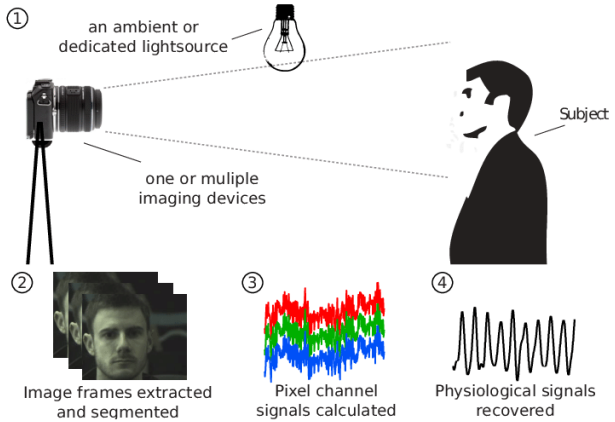
Mixed approach
Deep learning is just used for signal extraction (skin segmentation)



End-to-end deep learning
An end-to-end deep learning pipeline is used to directly map video stream to estimation of HR, HRV, etc.

Projects Proposals - rPPG

Heart Rate Monitoring



Wearables

ECG/PPG groundtruth

Nearables

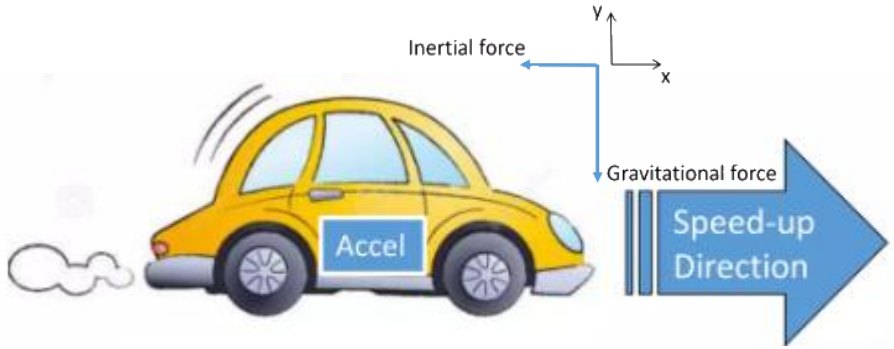
Remote
photoplethysmography

The main goals

- ❖ Acquisition of raw data;
- ❖ Machine learning/Deep learning models: Extract biomedical information from wearable and nearable sensors;
- ❖ Graphical User Interface to visualize the data and the extracted biomedical data.

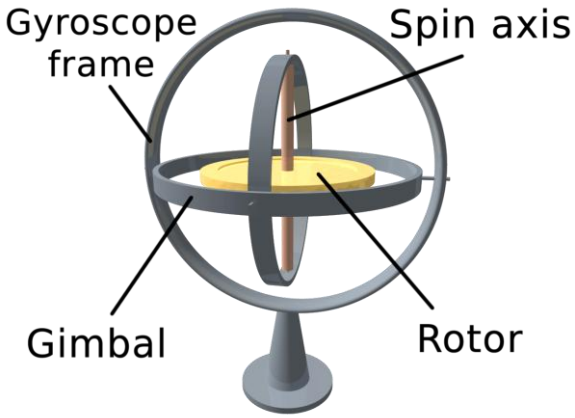
Projects Proposals – Pose Tracking

Accelerometer
3-axis: Acceleration

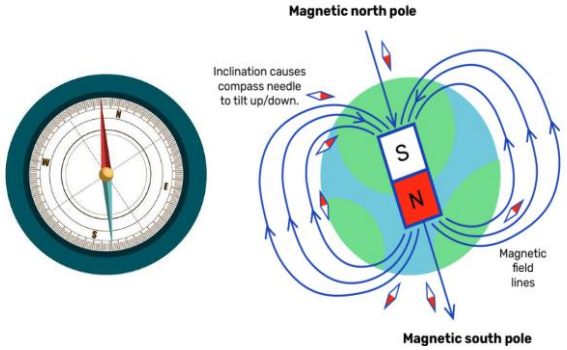


$$F = m \cdot a$$

Gyroscope
3-axis: Angular Speed



Magnetometer
3-axis: Magnetic field



IMU – Inertial Measurment Unit
- 6-axis: Accelerometer + Gyroscope
- 9-axis: Accelerometer + Gyroscope + Magnetometer

MEMS

Micro ElectroMechanical Systems

Projects Proposals – Pose Tracking

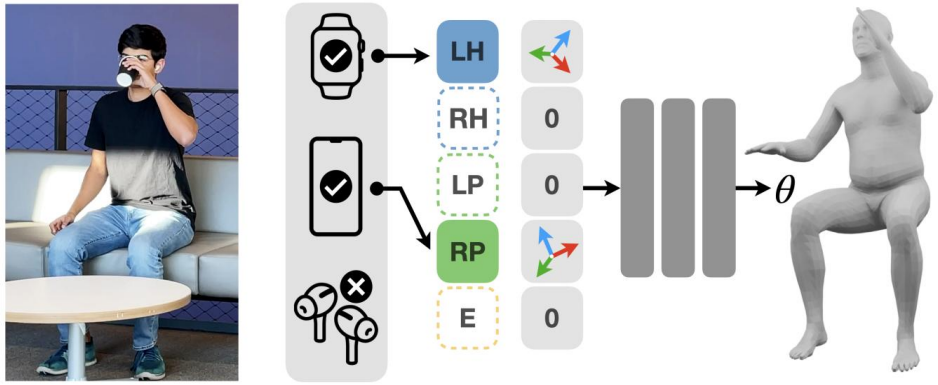
IMU Pose Estimation

Traditional method

$$
\begin{cases}
\dfrac{\partial \Delta \bar{\mathrm{R}}_{ij}}{\partial \mathbf{b}^g} = -\displaystyle\sum_{k=i}^{j-1} \left[\Delta \tilde{\mathrm{R}}_{k+1j}(\bar{\mathbf{b}}_i)^{\mathsf{T}}\, \mathrm{J}_r^k\, \Delta t\right] \\[2ex]
\dfrac{\partial \Delta \bar{\mathbf{v}}_{ij}}{\partial \mathbf{b}^a} = -\displaystyle\sum_{k=i}^{j-1} \Delta \tilde{\mathrm{R}}_{ij}(\bar{\mathbf{b}}_i)\Delta t \\[2ex]
\dfrac{\partial \Delta \bar{\mathbf{v}}_{ij}}{\partial \mathbf{b}^g} = -\displaystyle\sum_{k=i}^{j-1} \Delta \tilde{\mathrm{R}}_{ij}\left(\tilde{\mathbf{a}}_k - \bar{\mathbf{b}}_i^a\right)^{\wedge} \dfrac{\partial \Delta \bar{\mathrm{R}}_{ij}}{\partial \mathbf{b}^g} \Delta t \\[2ex]
\dfrac{\partial \Delta \bar{\mathbf{p}}_{ij}}{\partial \mathbf{b}^a} = -\displaystyle\sum_{k=i}^{j-1} \dfrac{3}{2} \Delta \tilde{\mathrm{R}}_{ij}(\bar{\mathbf{b}}_i)\Delta t^2 \\[2ex]
\dfrac{\partial \Delta \bar{\mathbf{p}}_{ij}}{\partial \mathbf{b}^g} = -\displaystyle\sum_{k=i}^{j-1} \dfrac{3}{2} \Delta \tilde{\mathrm{R}}_{ij}(\bar{\mathbf{b}}_i)\left(\tilde{\mathbf{a}}_k - \bar{\mathbf{b}}_i^a\right)^{\wedge} \dfrac{\partial \Delta \bar{\mathrm{R}}_{ij}}{\partial \mathbf{b}^g} \Delta t^2
\end{cases}
$$

Integrate equation to estimate pose

Deep learning method



Mollyn et al., IMUPoser: Full-Body Pose Estimation using IMUs in Phones, Watches, and Earbuds. 2023

End-to-end pose estimation

Video to IMUPoser

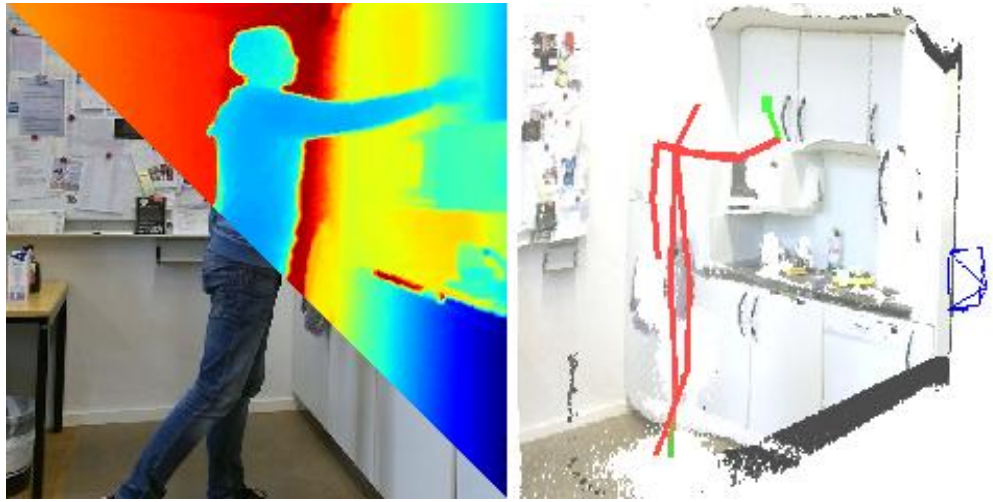
Projects Proposals – Pose Tracking

Camera human pose tracking

Bottom-up methods: estimate each body joint first and then group them to form a unique pose.
Top-down methods: run a person detector first and estimate body joints within the detected bounding boxes.

RGBd human pose tracking

Exploit depth information for 3d tracking



Projects Proposals – Pose Tracking

Dynamic Human Pose Estimation

Wearables

IMU

Nearables

Camera + pose estimation

The main goals

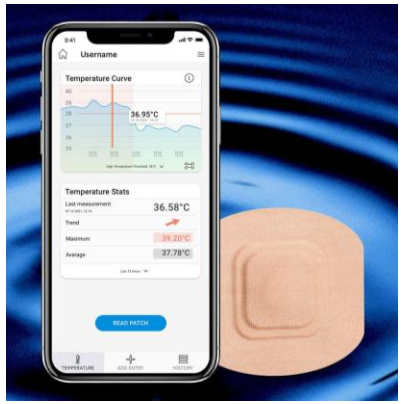
❖ Acquisition of raw data;

❖ Machine learning/ Deep learning models: Estimate body pose merging wearable and nearable data;

❖ Graphical User Interface to visualize the data and the processed dynamic human pose.

Projects Proposals – Thermotrack

Body Temperature Measurement



Wearables

Body temperature sensor

Nearables

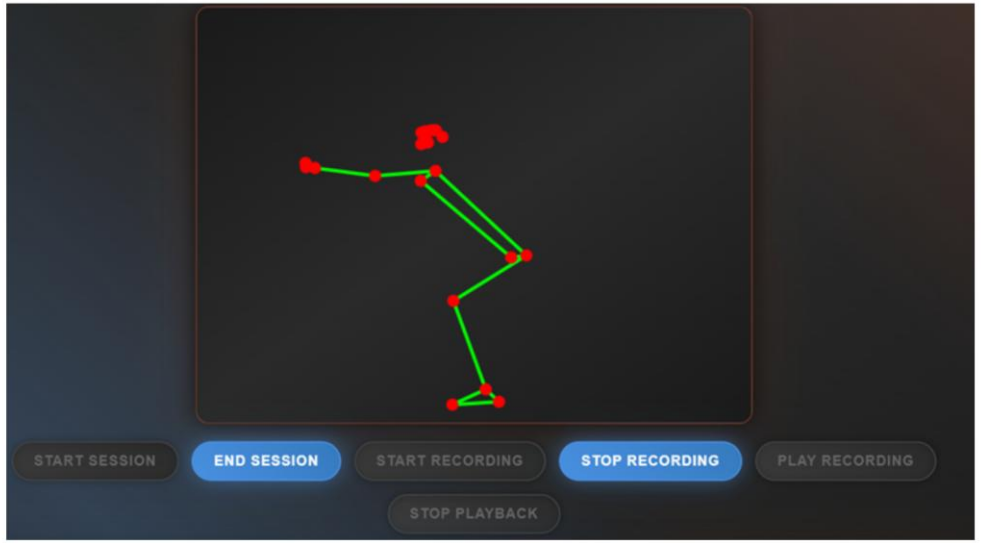
Infrared camera

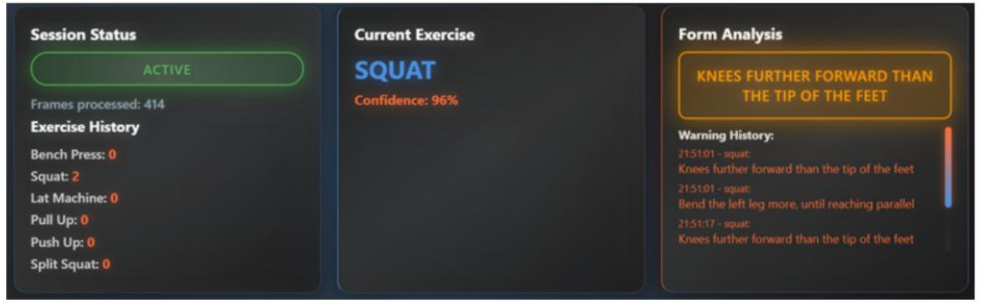
The main goals

- ❖ Acquisition of raw data;
- ❖ Machine learning/ Deep learning models: monitor variability of temperature using image processing and deep learning;
- ❖ Graphical User Interface to visualize the data and the processed body temperature.

Example Project

AI Gym Tracker





https://github.com/hannesvgel/gym_tracking

Project Guidelines

- ✓ **Groups of 3-4 students**
- ✓ **No group with only 1 or 2 students!**
- ✓ **Once the groups are defined, the students are required to submit a 1-page abstract (idea, goals, hardware requirements, selected software, and description of final demo).**

The development of the projects will start during the last part of the course under the guidance of the tutors. In case of need, it will be completed after course end, before the exam.

Course Program

Sessions

1. Introduction <- Today
2. Signal Processing
3. Image Processing
4. ML, DL for Data Science with Nearable Devices

Crash courses on algorithm tools you will use

5. Hardware/Software for Heart Rate Estimation
6. Hardware/Software for Temperature Analysis
7. Hardware/Software for Pose Estimation
8. Graphical User Interface (GUI)
9. GUI + Regulatory for AI-based Medical Devices

Focus on hardware/software tools for each project

10. - 13. Lab activities for Collaborative Projects development

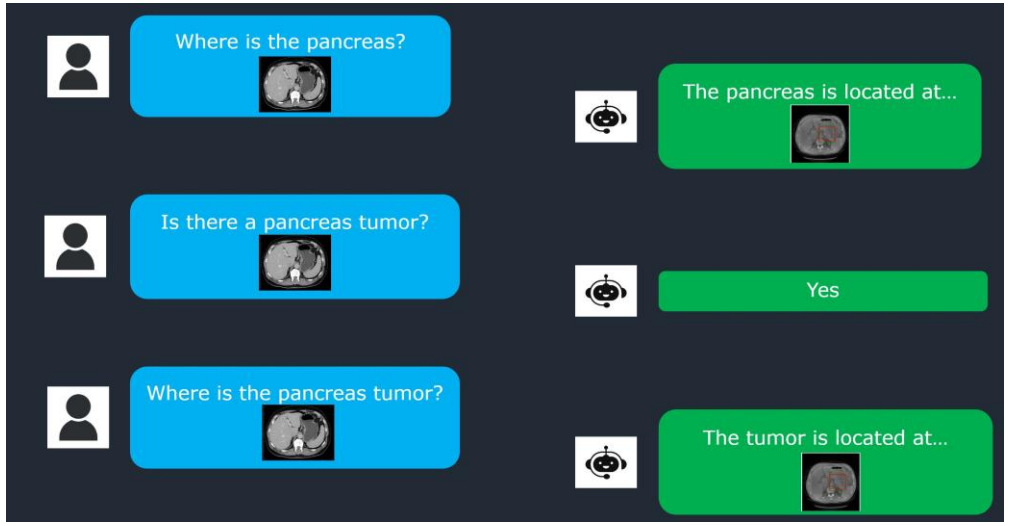
Time to develop your project

Thesis Proposals

AI Agent for Diagnosis of Pancreatic Cancer

Context: Pancreatic cancer reported the lowest 5-year survival rate among all cancer types in the United States in the surveilled period 2014-2020. A clinical issue is represented by small pancreatic cancers (<2 cm), which may sometimes escape the attention of experts looking at computed tomography (CT) scans.

Thesis aim: This thesis focuses on the use of artificial intelligence (AI) to support clinicians in the diagnosis of pancreatic cancer. More specifically, the goal is to develop an AI agent that can automatically detect small pancreatic cancers on CT.



Source: Moglia A, et al. J Healthc Inform Res (2025)

NVIDIA

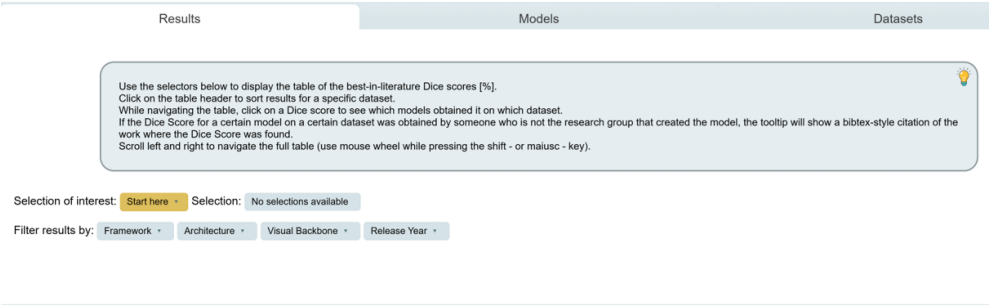
Partnership: Nvidia
Possible internship at the subsidiary in Poland

Contact: Prof: Andrea Moglia (andrea.moglia@polimi.it)

Thesis Proposals

AI Agent to Compare Models for Medical Image Segmentation

Context: The rapid advancement of generalist models in medical image segmentation has increasingly challenged specialist models, designed for a single task and trained on one or few datasets.
Thesis aim: This thesis focuses on the development of an AI agent capable of systematically comparing generalist and specialist models across different organs and datasets. The objective is to address the needs of computer scientists, engineers, and medical doctors seeking to know which approach yields superior performance, to support them in the selection of the most suitable one.



Source: Moglia A, et al. Inf Fus 2026. doi: 10.1016/j.inffus.2025.103709

https://hal9000-lab.github.io/GMMIS-Survey/

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